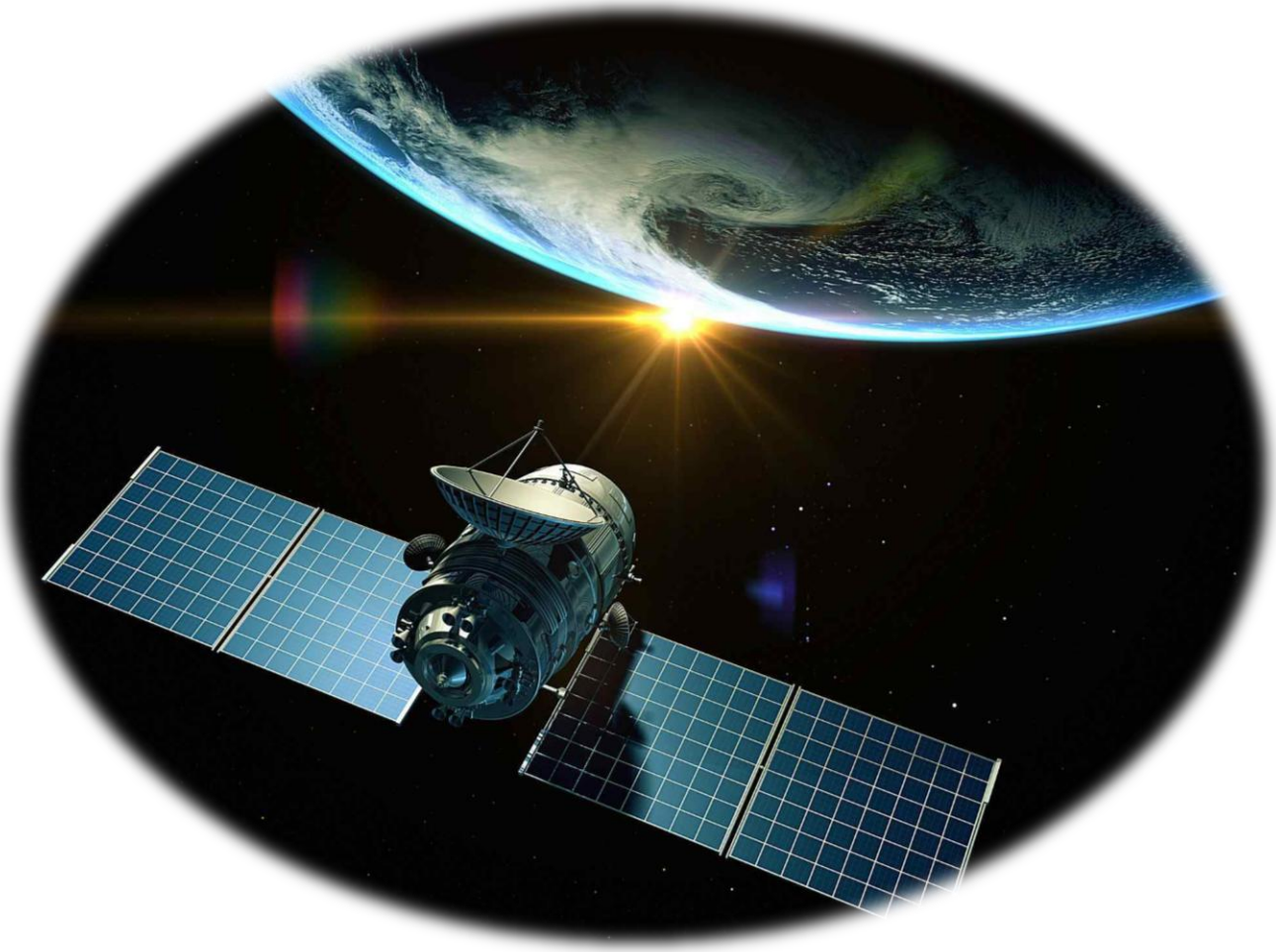
Basilisk Installation Guide on Windows



1.Introduction

Welcome to the Basilisk Installation Guide on Windows Operating System. This document provides a comprehensive, step-by-step walkthrough for installing and configuring the Basilisk simulation framework (version 2.2.1) on a Windows 10/11 machine. All instructions reflect the setup tested on Windows as of March 9, 2024, and are based on the official Basilisk documentation and GitHub resources. Whether you are new to Basilisk or upgrading from an earlier release, this guide will help ensure that your environment is correctly prepared and that the core components (Basilisk and Vizard) build and run without errors.

Requirements overview (hardware, OS version)

Before proceeding, please verify that your system meets the following prerequisites:

- **Operating System**: Windows 10 or Windows 11 (64-bit).
- **Hardware**: A multicore processor (Intel i5 or higher recommended), at least 8 GB of RAM, and ≥20 GB of free disk space.
- **Administrative Privileges**: You must be able to run Command Prompt (or PowerShell) as an administrator for certain installation steps (e.g., installing Visual Studio and running Conan).
- **Internet Access**: Required for downloading installers, Python packages, and Conan dependencies.

Software Requirements

- **CMake (≥ 3.14)**: Used to generate build files. Tested version: 3.29.0-rc2.
- **Python (≥ 3.8.x)**: Basilisk's Python interface requires Python 3. We recommend using Python 3.11.7. Ensure that "pip" is selected during installation.
- **pip**: Python's package installer. Confirm it was enabled at install time.
- **Visual Studio (15 2017 / 17 2022 or newer)**: Install the "Desktop development with C++" workload.
- **SWIG (version 3.x or 4.1; do not use 4.2)**: Required to generate Python bindings. Download and unzip SWIG (e.g., to C:\Program Files\Swig) so that swig.exe is reachable via your PATH.

Once you have confirmed that all of the above software and system requirements are in place, you are ready to move on to the detailed installation steps, beginning with setting up your environment variables and installing the necessary tools.

2.Software Setup

2.1 Install CMake

Download CMake

- Visit the official CMake website: https://cmake.org/
- Download a Windows installer for CMake version 3.14 or higher (the guide was tested on 3.29.0-rc2).

Run the Installer

- Launch the downloaded `.msi` and follow the prompts.
- When asked to select components, keep the defaults (including "Add CMake to the system PATH for all users" if offered).

Verify Installation

- Open a new Command Prompt (press **Win + R**, type cmd, and press **Enter**).
- Type: cmake –version
- You should see output similar to **cmake version 3.29.0-rc2**
- If you receive an error (e.g., "cmake is not recognized"), double-check that C:\Program Files\CMake\bin is on your PATH

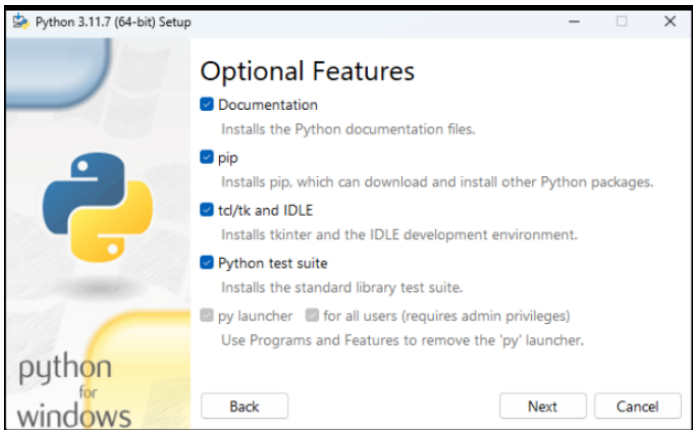
2.2. Install Python

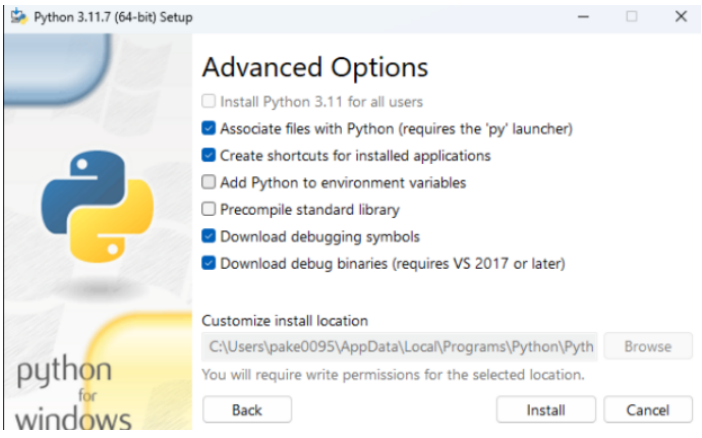
Download Python

- Go to the official Python downloads page: https://www.python.org/downloads/windows/
- Choose a Windows installer for Python 3.8.x or newer (this guide uses Python 3.11.7).

Run the Python Installer

- Launch the .exe installer.
- IMPORTANT: On the first screen, check the box “Add Python 3.x to PATH.”
- Click “Customize installation.”
- In the next screen, ensure “pip”, “tcl/tk and IDLE,” and “Documentation” are selected.
- Click “Next.”
- In the Advanced Options screen, check both:
- “Download debugging symbols”
- “Download debug binaries (requires VS 2017 or later).”
- Leave other defaults, and click “Install.”
- Wait for installation to complete, then click “Close.”





2.3. Install pip

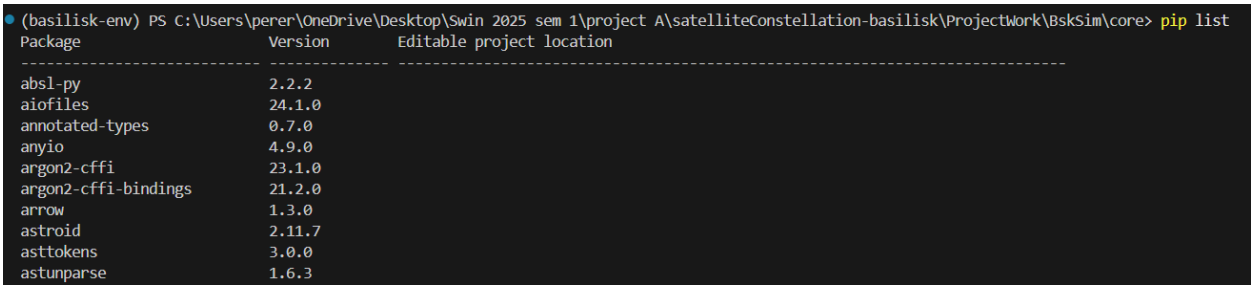
Verify Python & pip

- Open a new Command Prompt window.
- Type: **python –version**
- You should see something like: **Python 3.11.7**
- Next, verify pip: **pip –version**
- If either command fails, double-check that the installer added Python to your system PATH.

Confirm pip Installation

Note: If you selected "pip" during Python setup, no additional steps are needed here. However, we recommend running the following to ensure everything's in order.

- Upgrade pip (Optional but Recommended): **python -m pip install --upgrade pip**
- Test pip by Installing a Small Package: **pip install --upgrade setuptools**
- Check pip List: **pip list**
- Verify that setuptools and its dependencies appear in the list, this confirms that pip works correctly.



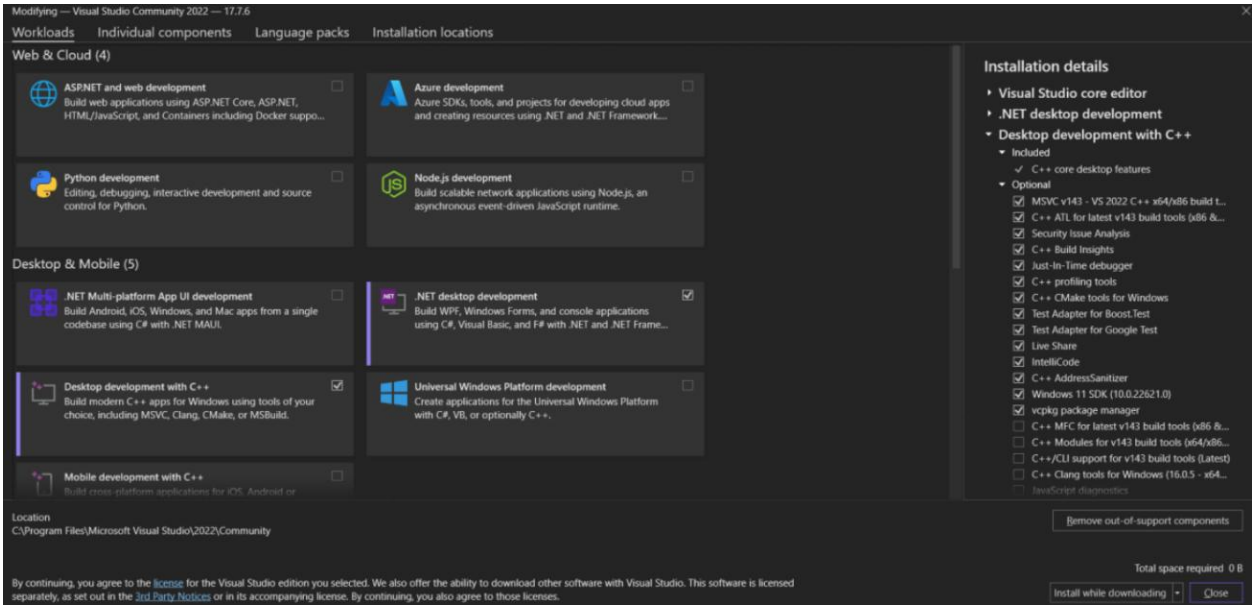
2.4. Install Visual Studio (C++)

Download Visual Studio Installer

- Go to https://visualstudio.microsoft.com/downloads/
- Download “Visual Studio 2022 Community”.

Run the Visual Studio Installer

- Launch the downloaded installer (vs_community.exe).
- In the “Workloads” tab, select “Desktop development with C++.” This ensures you have the MSVC compiler toolchain and Windows SDK.
- Optionally, under “Installation details” on the right, you can include:
  - “C++ CMake tools for Windows” (if it isn’t already implied by the workload).
  - “Windows 10 SDK” (10.0.x) (usually bundled).
- Click “Install” and wait for the components to download and install.



2.5. Install SWIG

Basilisk uses SWIG to generate Python bindings for its C++ modules.

Download SWIG

- Visit http://www.swig.org/download.html
- Under the "Windows" section, download **"swigwin-4.1.1.zip"** (or any SWIG 4.1.x).
- **DO NOT** use SWIG 4.2, as it is known to cause build issues on Windows.

Unzip SWIG

- Extract the contents of the ZIP (which contains swig.exe and support files) to a permanent location, e.g.: **C:\Program Files\Swig**
- The final path should contain swig.exe, e.g.: **C:\Program Files\Swig\swig.exe**

Verify SWIG Installation

- Open a new Command Prompt.
- Type: **swig -version**
- You should see output similar to: **SWIG Version 4.1.1**

If the command fails, you will add SWIG's directory to your PATH in the next subsection.

2.6. Configure Environment Variables

Several tools installed above need to be discoverable via the command line. This subsection walks you through setting system (or user) environment variables so that cmake, swig, and later Basilisk binaries can be found.

Important: Whenever you modify environment variables, you must close and reopen any open Command Prompt windows and sometimes reboot to ensure changes take effect.

Open Environment Variables Panel

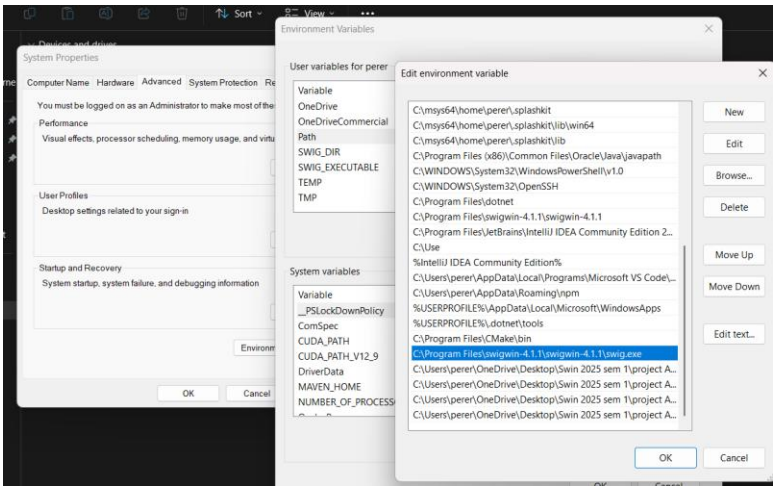
- Right-click **This PC** (or **My Computer**) on your Desktop or in File Explorer, then click **"Properties."**
- In the System window, click **"Advanced system settings"** on the left.
- In the "System Properties" dialog, under the **Advanced** tab, click **"Environment Variables..."**

Add CMake to PATH (if not already added)

- In the **User variables** (or **System variables**) section, locate and select **"Path,"** then click **"Edit."**
- Click **"New"** and enter (or verify) the folder where cmake.exe lives. By default, that is: **C:\Program Files\CMake\bin**
- Click **"OK"** to save.

Add SWIG to PATH

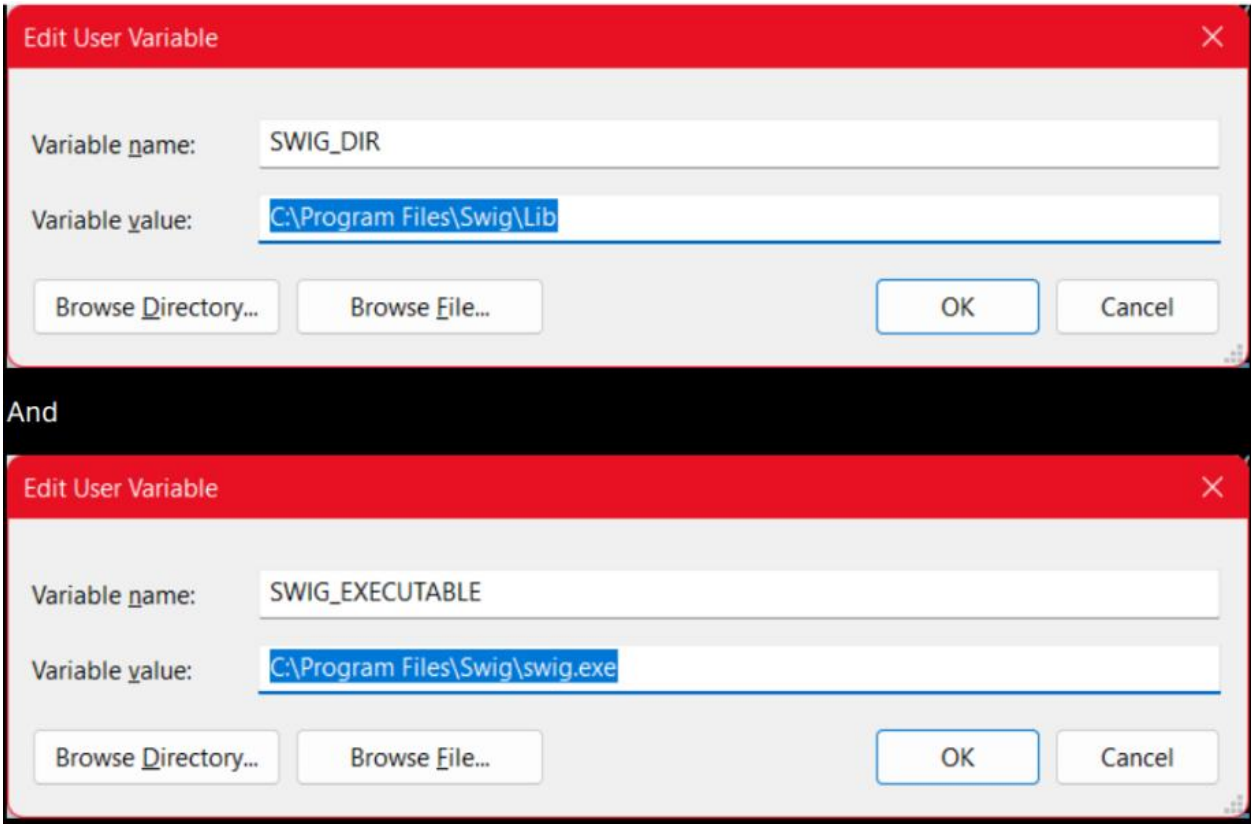
- In the same **Path** editor (under User or System variables), click **"New"** and add: **C:\Program Files\Swig**
- Click **"OK."**



Create SWIG_DIR and SWIG_EXECUTABLE Variables

- Under **User variables** (or **System variables**), click **"New."**
  - For **Variable name**, enter: **SWIG_DIR**
  - For **Variable value**, enter the swig root folder (where swig.exe resides), e.g.: **C:\Program Files\Swig**
- Click **"OK."**

- Again, click **"New."**
  - For **Variable name**, enter: **SWIG_EXECUTABLE**
  - For **Variable value**, enter the full path to swig.exe, e.g.: **C:\Program Files\Swig\swig.exe**
- Click **"OK."**



3. Download Basilisk

Instead of downloading a ZIP archive, Basilisk must be cloned using Git to ensure all submodules and version information are correctly retrieved. Follow the steps below to clone the v2.2.1

3.2. Clone the Basilisk Repository

Open Command Prompt

- Navigate to the directory where you want to keep the Basilisk source (e.g., C:\Dev\, D:\Projects\, etc.):

Clone the Repository

- Run: git clone https://github.com/AVSLab/basilisk.git
- This creates a folder named basilisk.

Note: Go to https://github.com/AVSLab/basilisk/tags and find the correct basilisk version(2.2.1).

4. Python Environment Setup

This section guides you through creating an isolated Python environment for Basilisk, installing all required packages, and verifying the installation.

4.1. Install virtualenv

Open a Command Prompt

- Press Win + R, type cmd, and press Enter.
- Navigate to your Basilisk root directory (e.g., where you cloned basilisk)

Install virtualenv Globally

- Run: **pip install virtualenv**
- Wait for pip to download and install virtualenv.

4.2. Create a Virtual Environment

- From the Basilisk Root Folder, create a new virtual environment named basilisk-env by entering: **virtualenv basilisk-env**
- This command creates a basilisk-env subfolder containing an isolated Python interpreter and package directory.

Confirm Creation

- Verify that a new folder named basilisk-env exists alongside folders such as basilisk, src, and examples.

Activate the Virtual Environment

- In your Command Prompt (still in C:\Dev\basilisk), run: **.\basilisk-env\Scripts\activate.bat**
- After activation, your prompt should prepend with (basilisk-env), indicating all python and pip commands now use this isolated environment.

4.3. Install Required Python Packages

- Within the Activated Environment ((basilisk-env)), install Basilisk's core Python dependencies by running: **pip install pandas numpy matplotlib pytest Pillow conan==1.63.0 parse>=1.18.0**
- Additional dependencies (e.g., scipy, networkx, pyyaml) will be fetched automatically when you build Basilisk with Conan.

4.4. Verify Installed Packages

List Installed Packages

- Still in (basilisk-env), run: **pip list**
- Confirm that at least the following entries appear:

```
astroid==2.11.7
bottle==0.12.25
certifi==2024.2.2
charset-
normalizer==3.3.2
colorama==0.4.6
conan==1.63.0
contourpy==1.2.0
cycler==0.12.1
deprecation==2.0.7
dill==0.3.8
distro==1.1.0
eigen==0.0.1
fasteners==0.19
fonttools==4.49.0
future==0.16.0
idna==3.6
iniconfig==2.0.0
isort==5.13.2
Jinja2==3.1.3
kiwisolver==1.4.5
lazy-object-
proxy==1.10.0
MarkupSafe==2.0.1
matplotlib==3.8.3
mccabe==0.7.0
node-semver==0.6.1
numpy==1.26.4
packaging==23.2
pandas==2.2.1
parse==1.20.1
patch==1.16
patch-ng==1.17.4
pillow==10.2.0
platformdirs==4.2.0
pluggy==1.4.0
pluginbase==0.7
Pygments==2.17.2
PyJWT==2.8.0
pylint==2.14.2
pyparsing==3.1.1
pytest==8.0.2
python-
dateutil==2.9.0.pos
t0
pytz==2024.1
PyYAML==6.0.1
requests==2.31.0
six==1.16.0
tomlkit==0.12.4
tqdm==4.66.2
tzdata==2024.1
urllib3==1.26.18
wrapt==1.16.0
```

• Run pip list and compare against the requirements.txt list. If some packages are missing you have to install them manually.

5. Build Basilisk with Conan

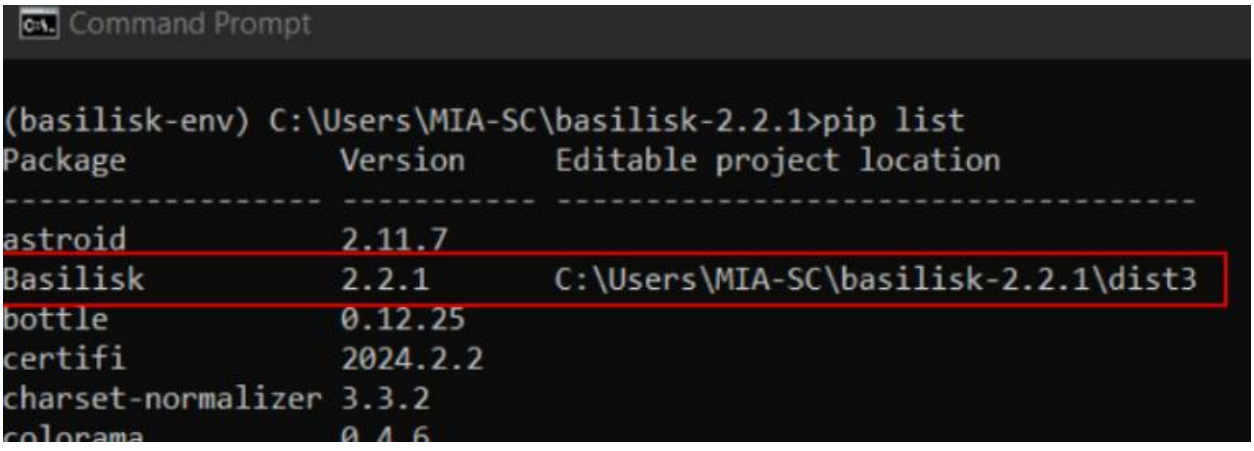
5.2. Run Conan Install

Note: You have to close all the terminals and reopen it with activated environment

- Click Start, type “cmd”, then right-click “Command Prompt” and select “Run as administrator.”
- Execute the Conan build command: **python conanfile.py --generator "Visual Studio 17 2022"**
- **Note**: The build can take **10–15 minutes** on typical hardware. Remain patient and watch for any error messages.

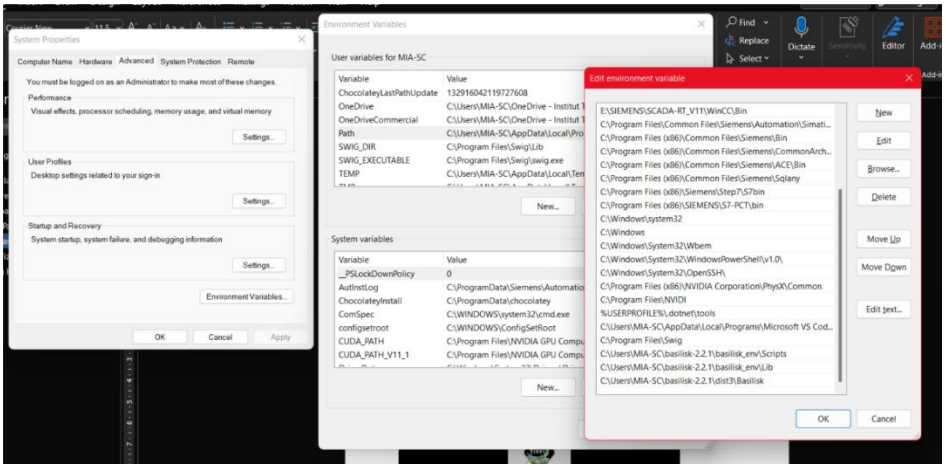
5.3. Verify Basilisk Package

- After Conan completes without errors, open a **new** (non-Administrator) Command Prompt.
- Reactivate your virtual environment and return to the Basilisk root.
- List installed Python packages to confirm that **basilisk** appears.
- You should see an entry similar to:



- Confirm that a new dist3 folder exists inside your Basilisk root.
- The presence of **dist3\Basilisk** indicates that the C++ library and Python module were built and staged correctly.

- Add a new environment variable path: C:\Users\MIA-SC\basilisk-2.2.1\dist3\Basilisk



6. Run basic Basilisk example code

- First, open a Command Prompt and navigate to your Basilisk installation directory. Activate the virtual environment you created earlier by running.
- Once activated, change into the examples subfolder: **cd examples**
- To execute a simple, non-visual example that generates orbital plots, enter: **python scenarioBasicOrbit.py**
- This script will produce a series of Matplotlib windows displaying various orbital state variables (e.g., position, attitude). Simply close the plot windows when you are finished reviewing them.

7. Install Vizard for 3D Visualization

Vizard provides real-time 3D visualization of Basilisk simulations. Follow the steps below to download, extract, and verify Vizard on Windows.

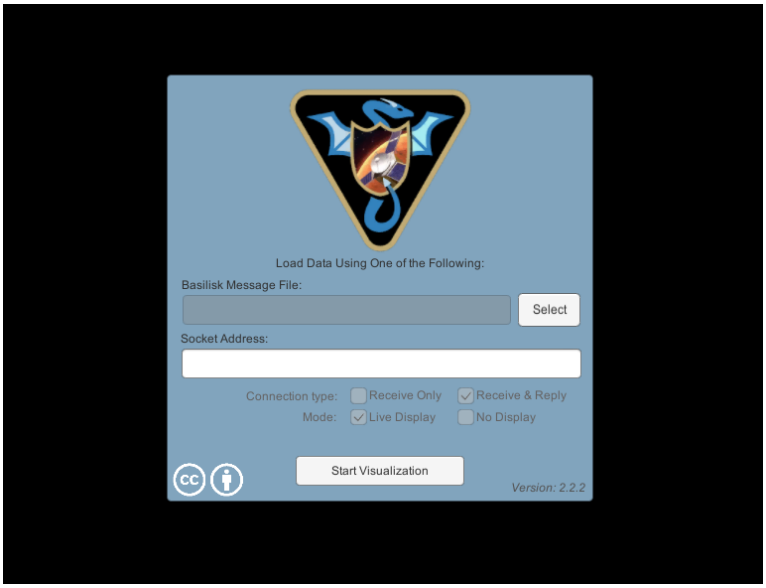
7.1. Download Vizard

Open Your Web Browser and navigate to:
https://hanspeterschaub.info/basilisk/Vizard/VizardDownload.html

Download the ZIP Archive to a folder of your choice

Extract and Run Vizard

- Navigate to the Download Location in File Explorer (e.g., C:\Downloads).
- Right-Click the ZIP Archive (Vizard-1.0-win.zip) and select **"Extract All..."**.
- Inside the extracted Vizard folder, go to Windows\Vizard and run the Vizard.exe file
- If it is working, a new window will be shown



7.2. Run the example of VIZARD live visualization:

- Go to Basilisk example directory: cd \basilisk-2.2.1\examples
- Run the example: python scenarioBasicOrbitStream.py
- Copy the tcp address: **tcp://localhost:5556**
- Paste to Vizard socket address:

Troubleshooting & Tips

Common build errors (e.g., missing environment variables, incompatible SWIG version)

Verifying correct Python interpreter and virtualenv activation

Appendices

1). Full list of required Python packages (requirements.txt)